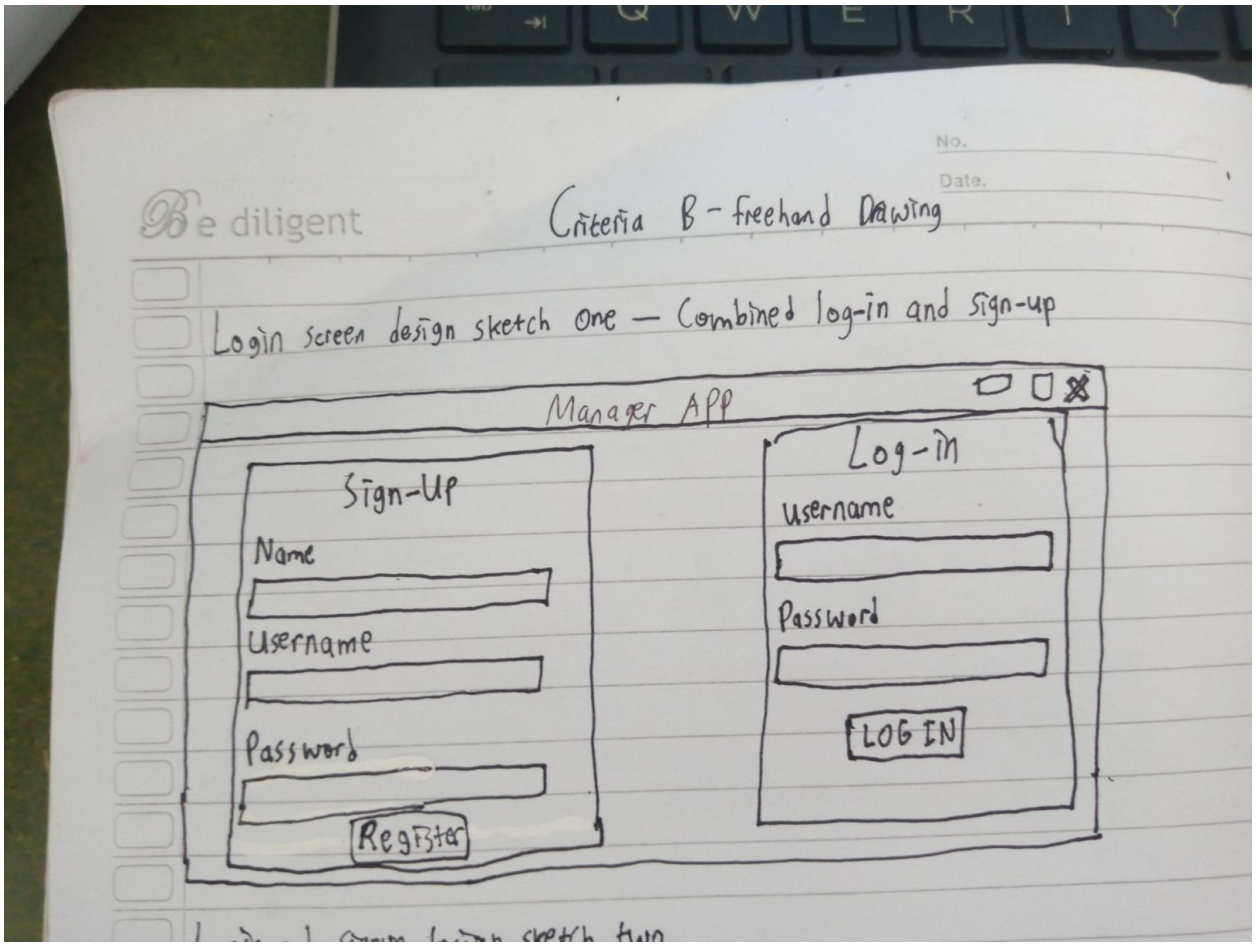
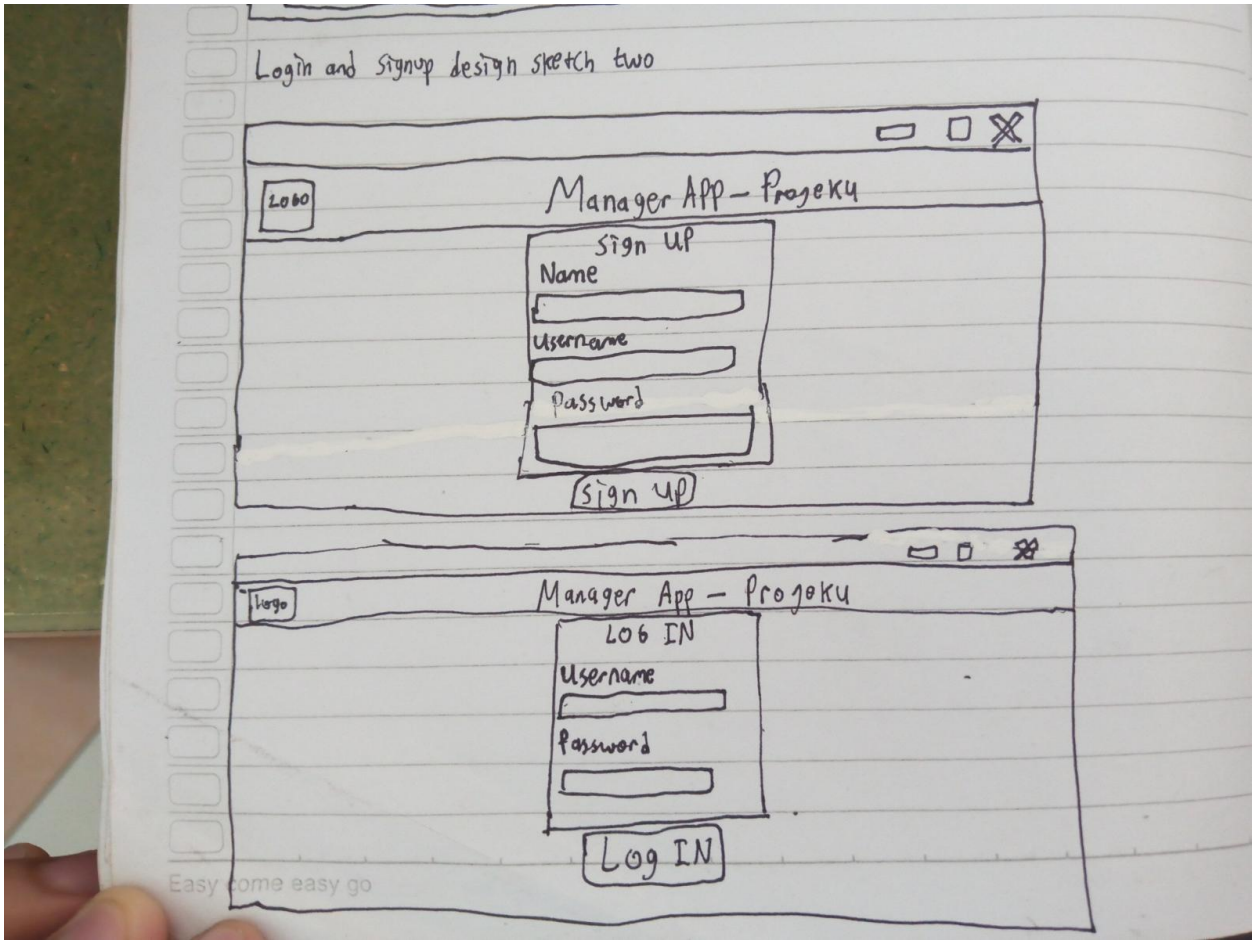
I. Initial Designs (sketch by hand)

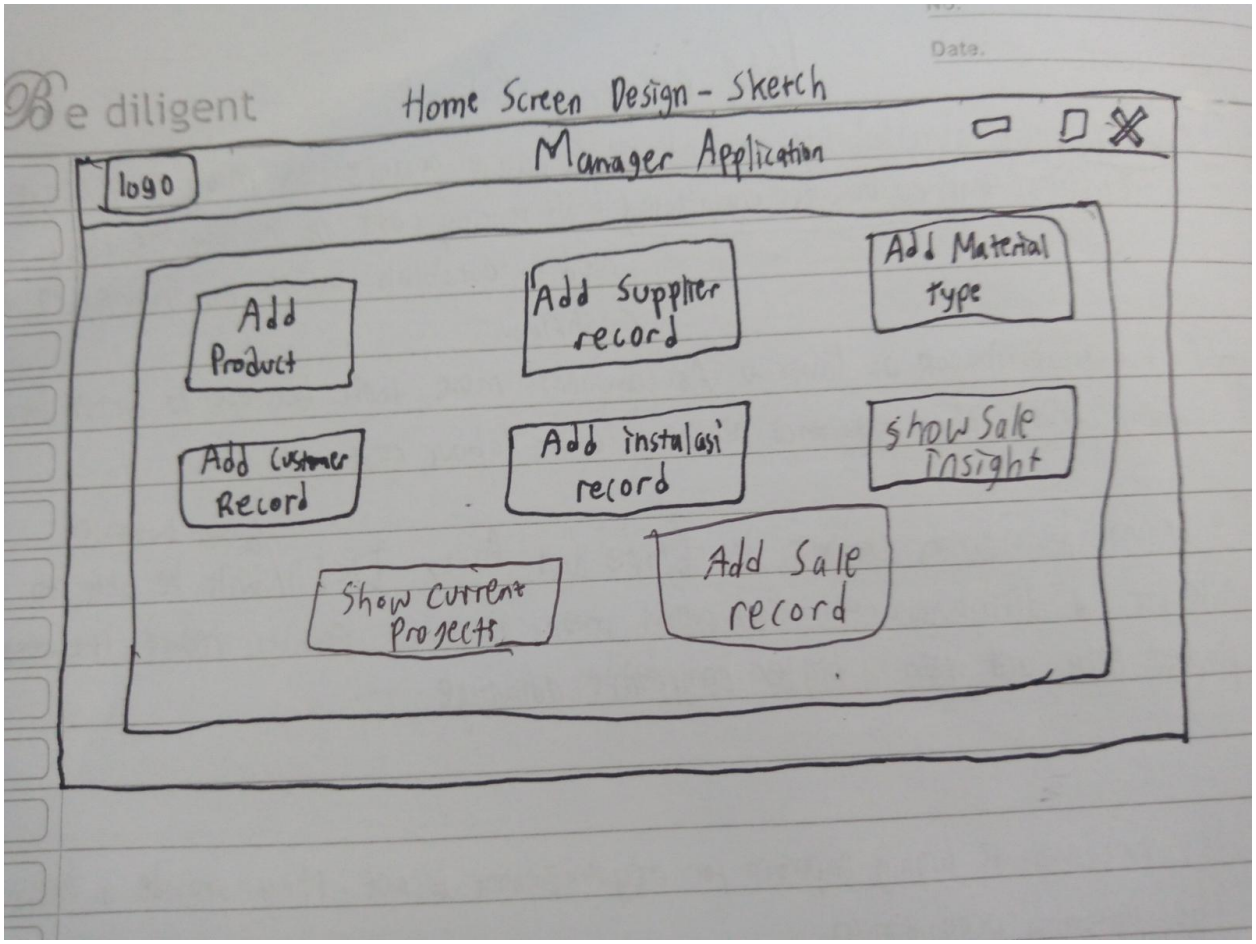
Sign up and Login screen sketch - design one



Sign up and Login screen sketch - design one

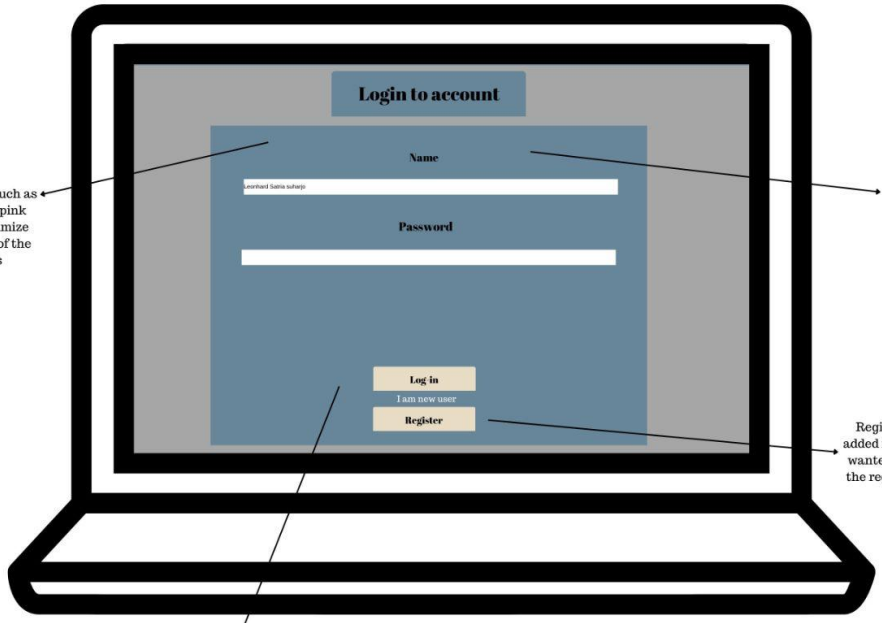


Home Screen Design sketch =>



II. Screen Designs

Compsci IA App Login Page - Design Two



Vibrant Colors such as navy blue and pink chosen to maximize comfortability of the user's eyes

Use of floating box, to maximize simplicity and function, as what the user needs and wants

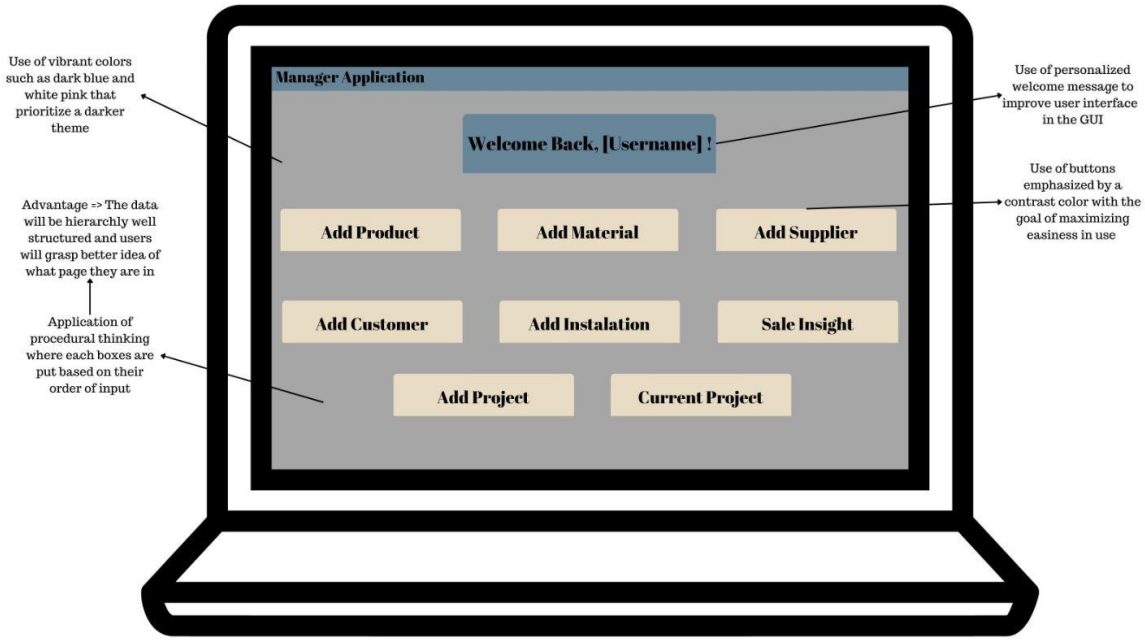
Register button is added in case the user wanted to go back to the registration page

The Registration and login pages are seperated, to increase clarity and reduce load on user

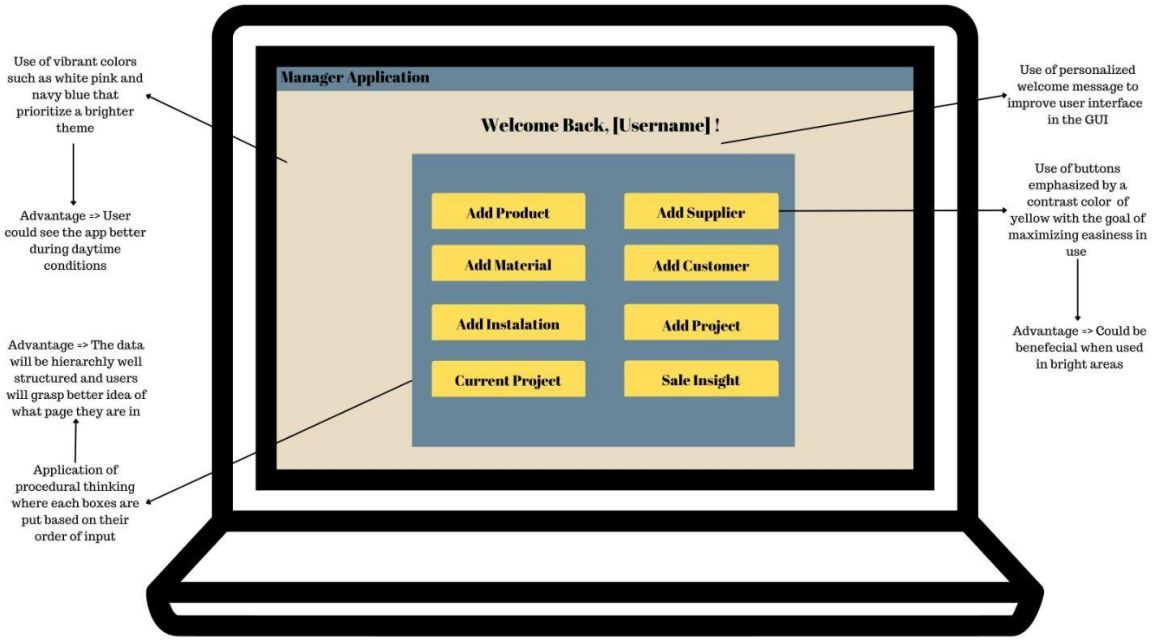
Advantage => user will feel less overwhelmed when using the app since they know exactly what they are doing in the page

Disadvantage => User might found it ineffecient to have two different pages, where they should click the button to move to the next/previous page

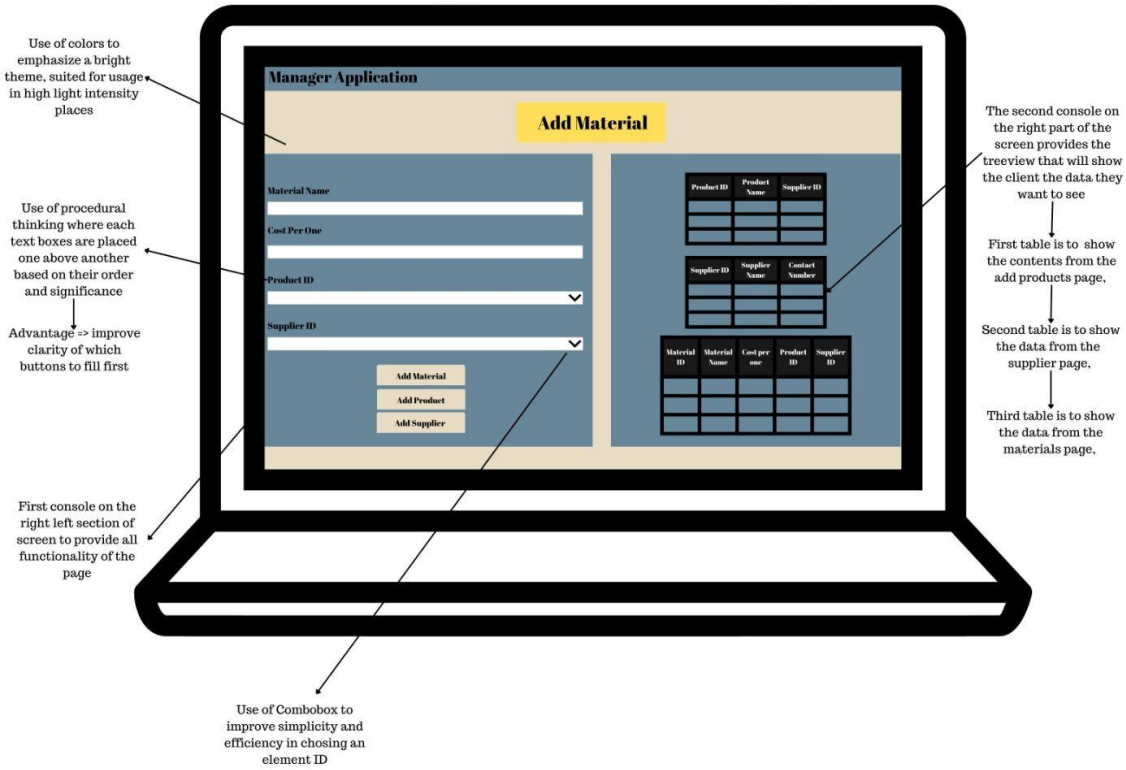
Compsci IA app - Home page design one



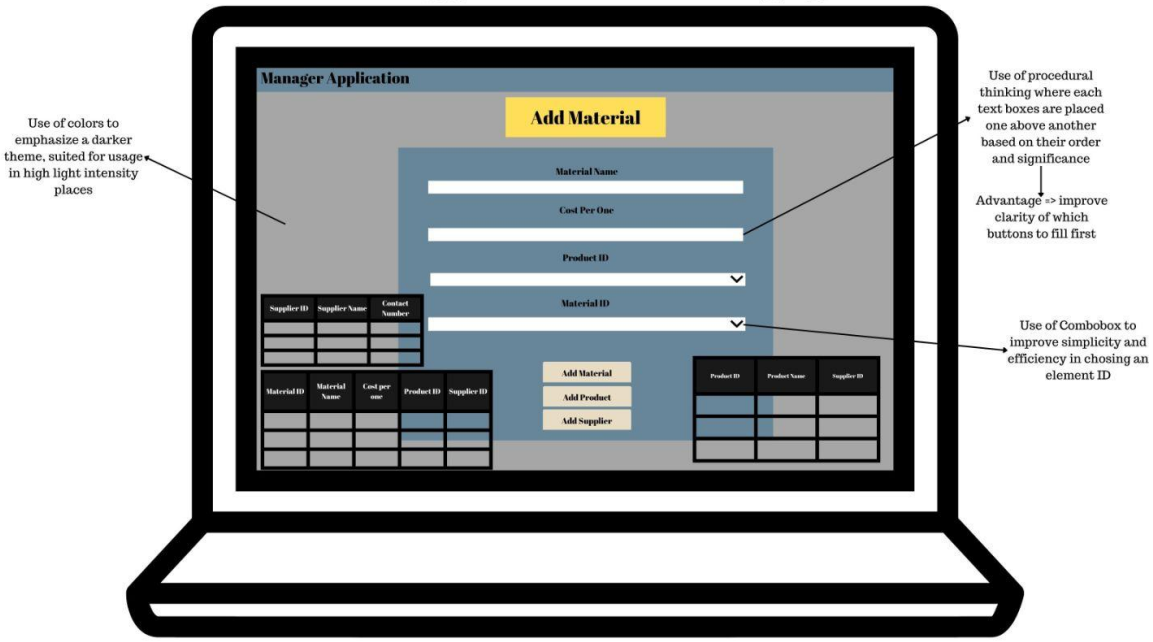
Compsci IA app - Home page design two



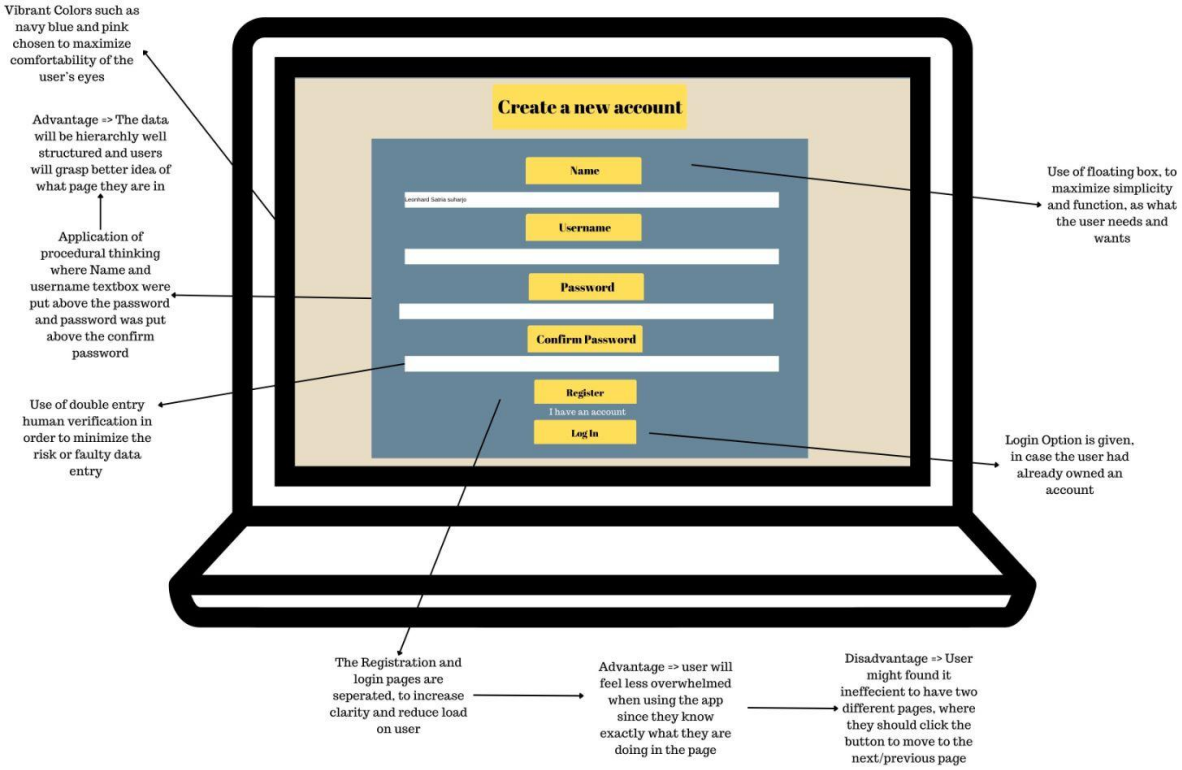
Compsci IA app - Sub pages theme design one
*To be applied to other sub pages



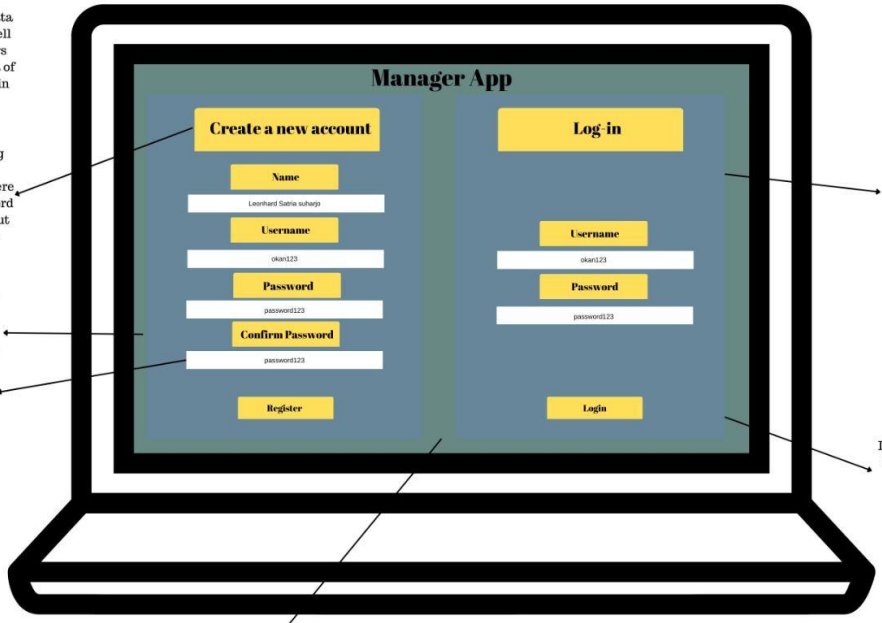
Compsci IA app - Sub pages theme design two
*To be applied to other sub pages



Compsci IA App Registration Page - Design Two



Login and registration Design one



Advantage => The data will be hierarchly well structured and users will grasp better idea of what page they are in

Application of procedural thinking where Name and username textbox were put above the password and password was put above the confirm password

Vibrant color of navy blue variants are chosen to give a comfortable one tone color

Use of double entry human verification in order to minimize the risk or faulty data entry

Use of floating box, to maximize simplicity and function, as what the user needs and wants

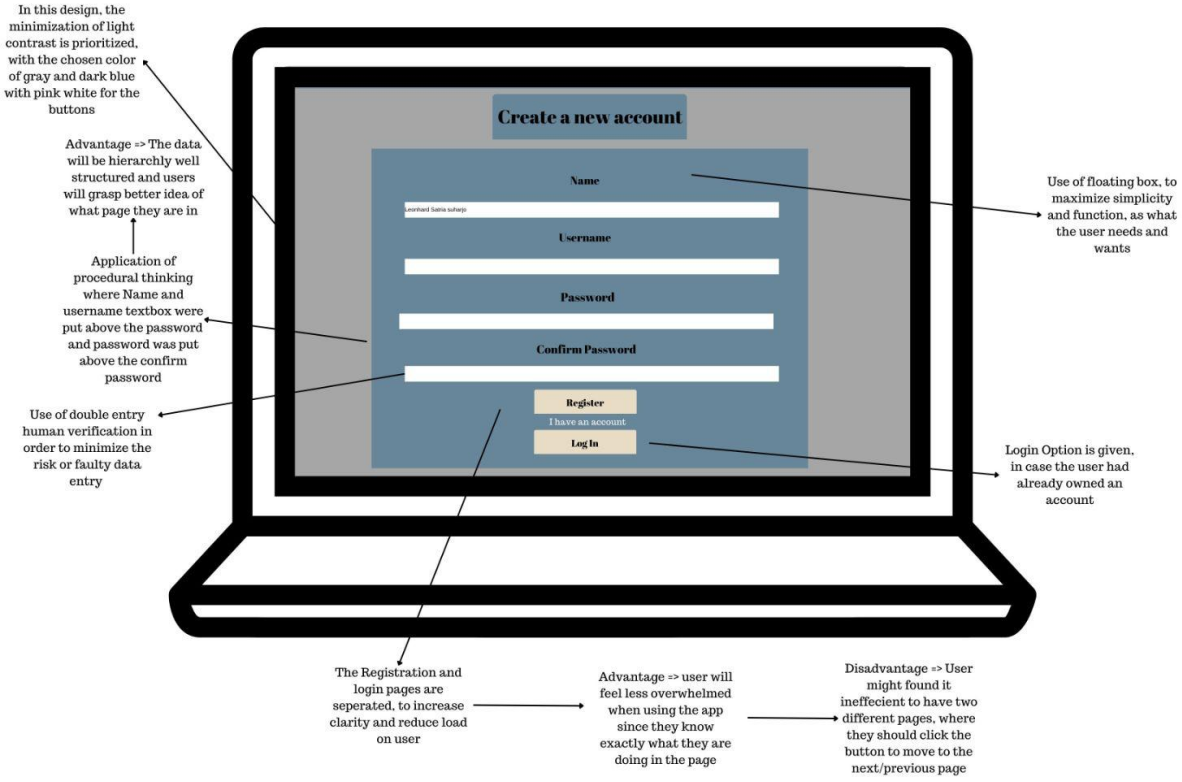
Login Option is given, in case the user had already owned an account

Registration and login sections are combined in one page to increase access effeciency

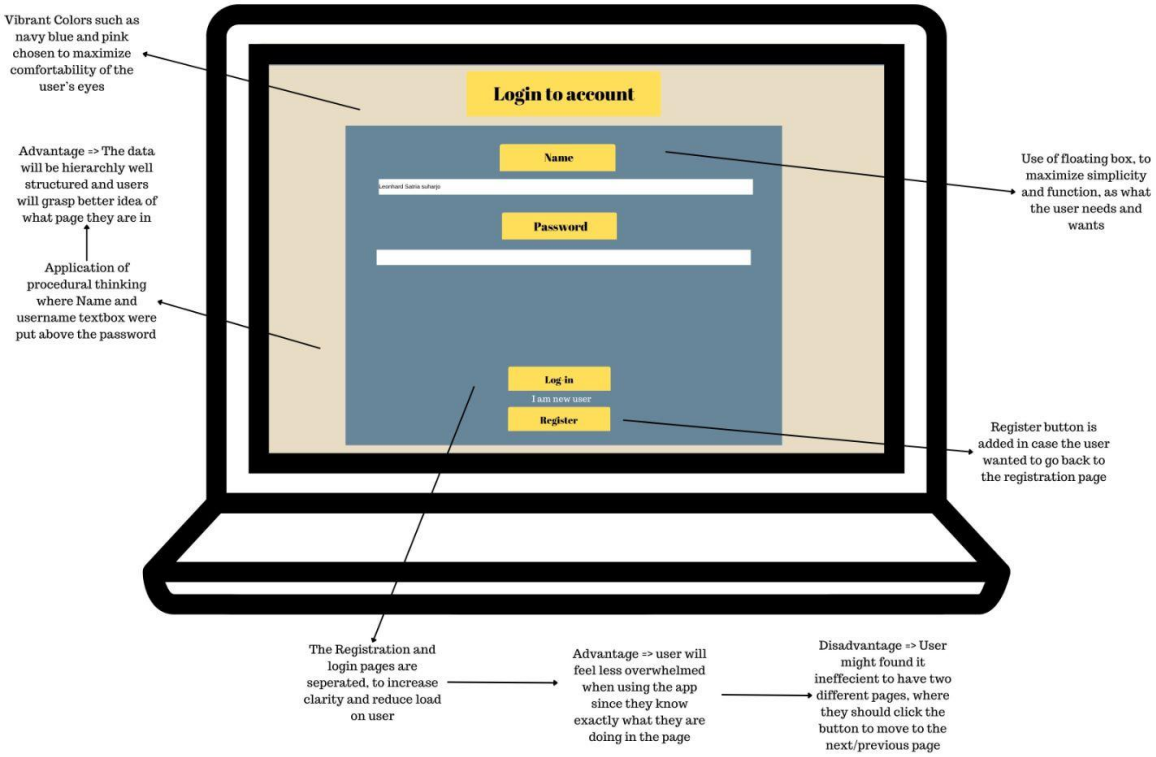
Advantage => user will experience easier process since both of them are directly located in one page

Disadvantage => User might feel overwhelmed and anxiety when dealing with a lot of information in one screen

Compsci IA App Registration Page - Design Three



Compsci IA App Login Page - Design Two



III. Data Dictionary

Add Product Page						
Field Name	Data Type	PK/FK	Field Size	Verification	Validation	Description
Product ID	integer	PK (auto)	10000 bytes	-	-	Serves as the primary key for the product table in the database
Product_Name	text	-	1000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for each product

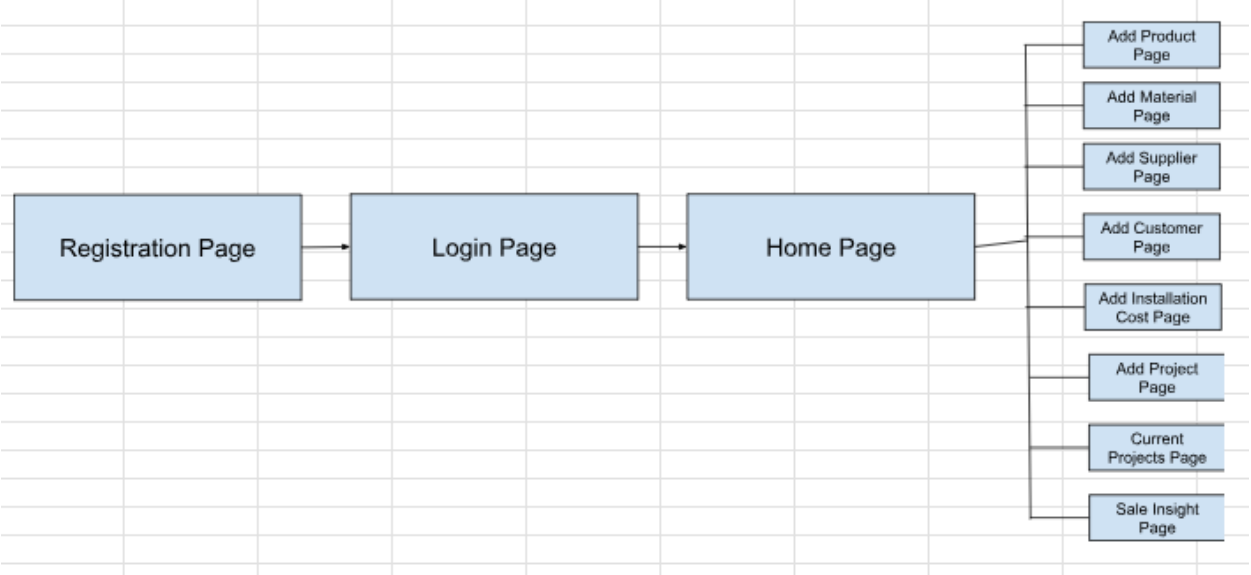
Add Material Page						
Field Name	Data Type	PK/FK	Field Size	Verification	Validation	Description
Material ID	integer	PK (auto)	10000 bytes	-	-	Serves as the primary key for the material table in the database
Material_name	text	-	1000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for each material
Price per one	integer	-	10000 bytes	-	-	Serves as the number that represents
Product ID	Integer	FK (PK in product Table)	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for product to be able to connect each material with each product
Supplier_ID	Text	FK (PK in supplier Table)	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for supplier to be able to connect each material with the supplier page

Add Supplier Page						
Field Name	Data Type	PK/FK	Field Size	Verification	Validation	Description
Supplier ID	integer	PK (auto)	10000 bytes	-	-	Serves as the primary key for the material table in the database
Supplier_Name	Text	-	1000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for each supplier
Contact_NO	integer	-	10000 bytes	-	-	Serves as the number that represents the contact number of each supplier
Supplier_status	Text	-	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the text that represents that the status of the supplier either active or unactive
Address	Text	-	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the text that shows the location address of the supplier

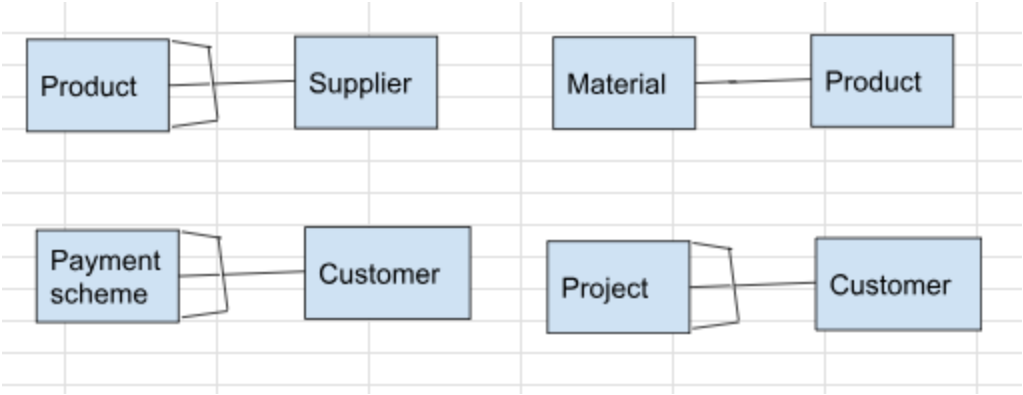
Add Customer Page						
Field Name	Data Type	PK/FK	Field Size	Verification	Validation	Description
Customer ID	integer	PK (auto)	10000 bytes	-	-	Serves as the primary key for the customer table in the database
Customer_Name	Text	-	1000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for each customer
Customer_contact_NO	integer	-	10000 bytes	-	-	Serves as the number that represents the contact number of each customer
Address	Text	-	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the text that shows the location address of the customer

Project page						
Field Name	Data Type	PK/FK	Field Size	Verification	Validation	Description
Summary ID	integer	PK (auto)	10000 bytes	-	-	Serves as the primary key for the project table in the database
Project_Name	Text	-	1000 bytes	Visual check	Presence check, Data type check, length check	Serves as the identifier for each project
Project Value	integer	-	10000 bytes	-	-	Serves as the number that represents the total value of the project
Installation expense	integer	-	10000 bytes	-	-	Serves as the number that represents the installation cost of the project
Material Expense	Text	-	10000 bytes	Visual check	Presence check, Data type check, length check	Serves as the text that shows the material cost of the project
Gross Profit	Float	-	10000000 bytes	-	-	Serves as the text that shows the gross profit of the project

IV. General Data overview

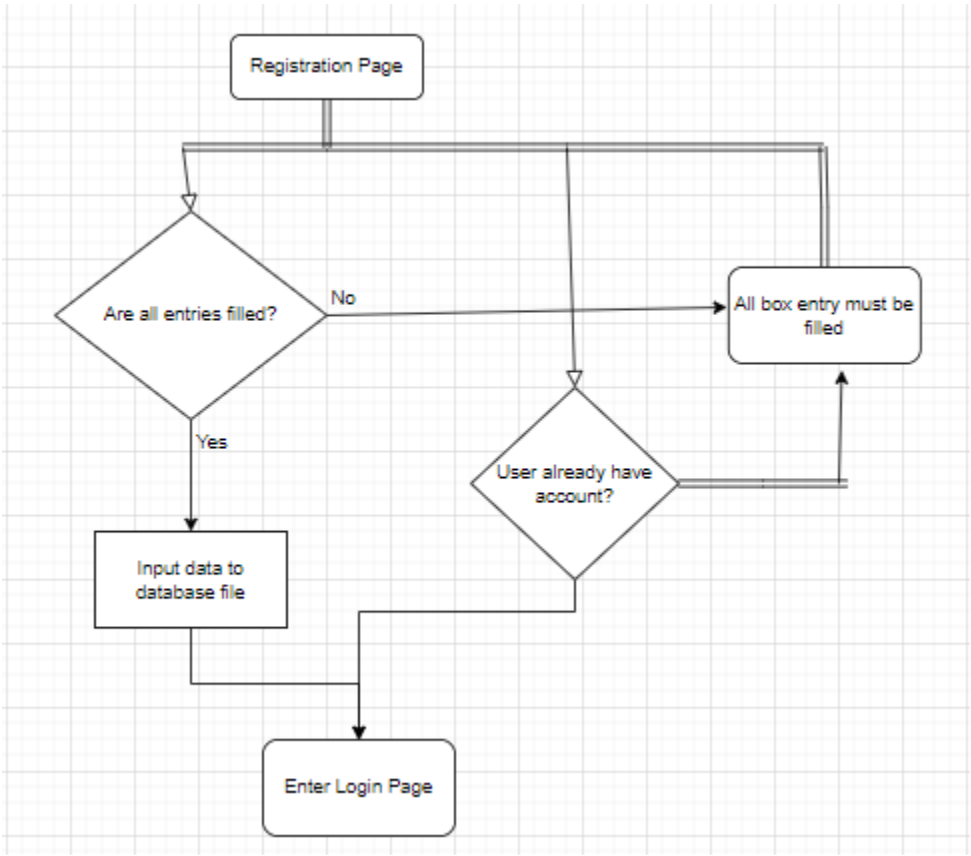


V. ERD

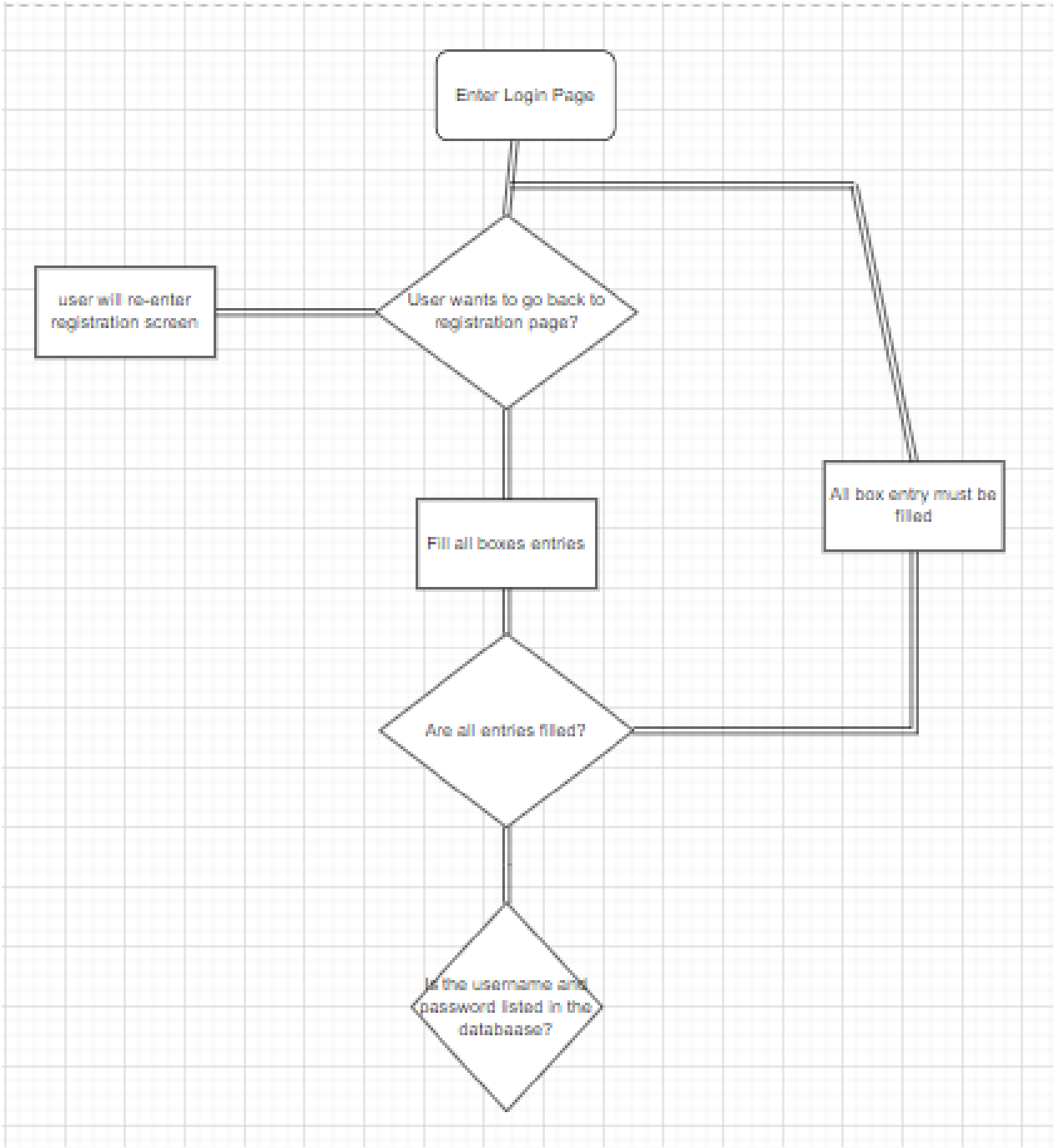


VI. Flowchart

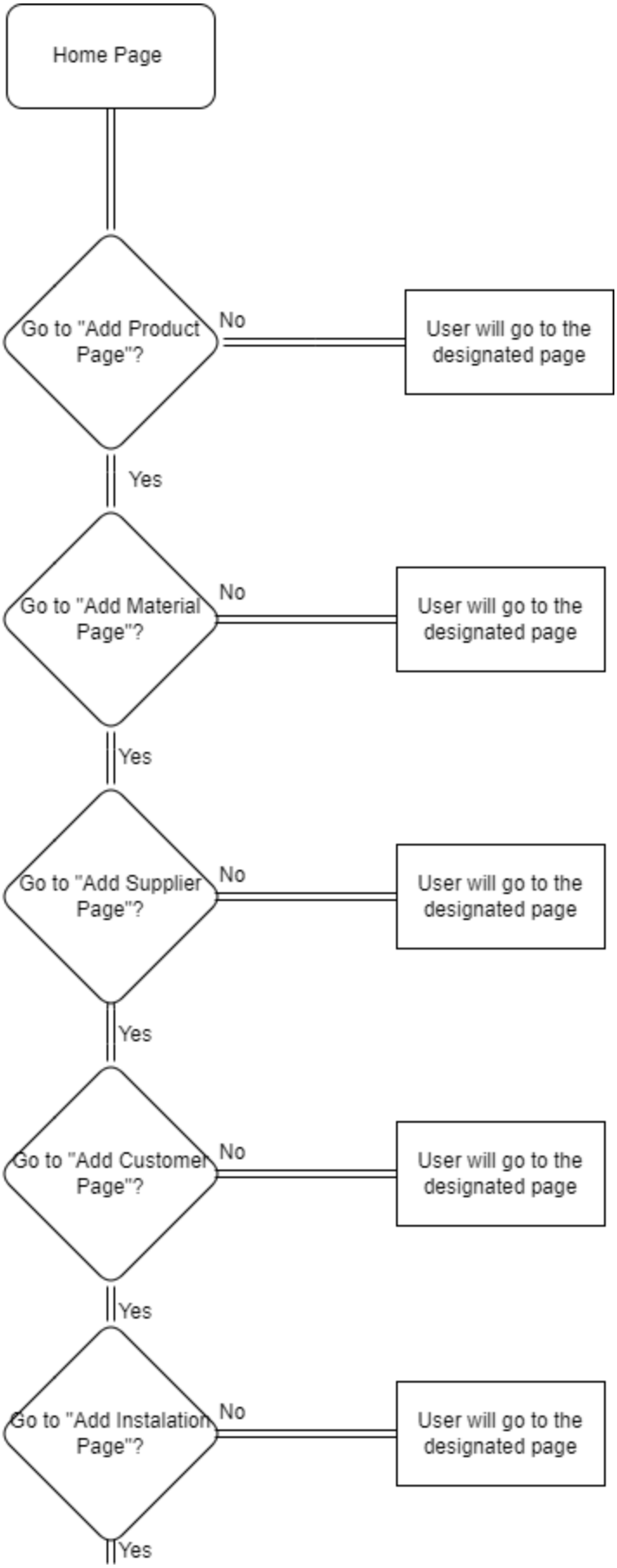
Registration screen flowchart

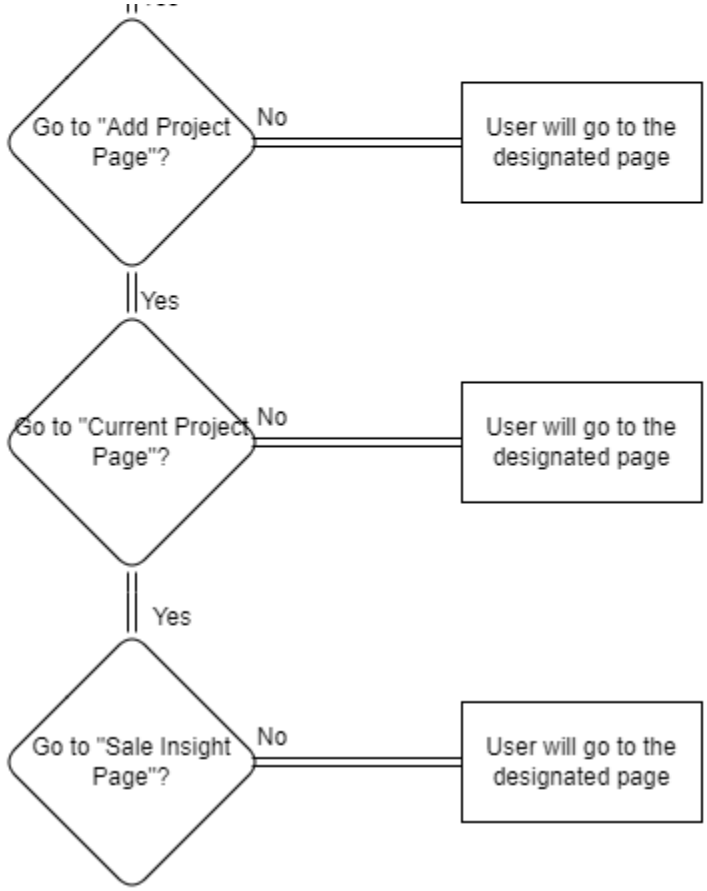


Login screen flowchart

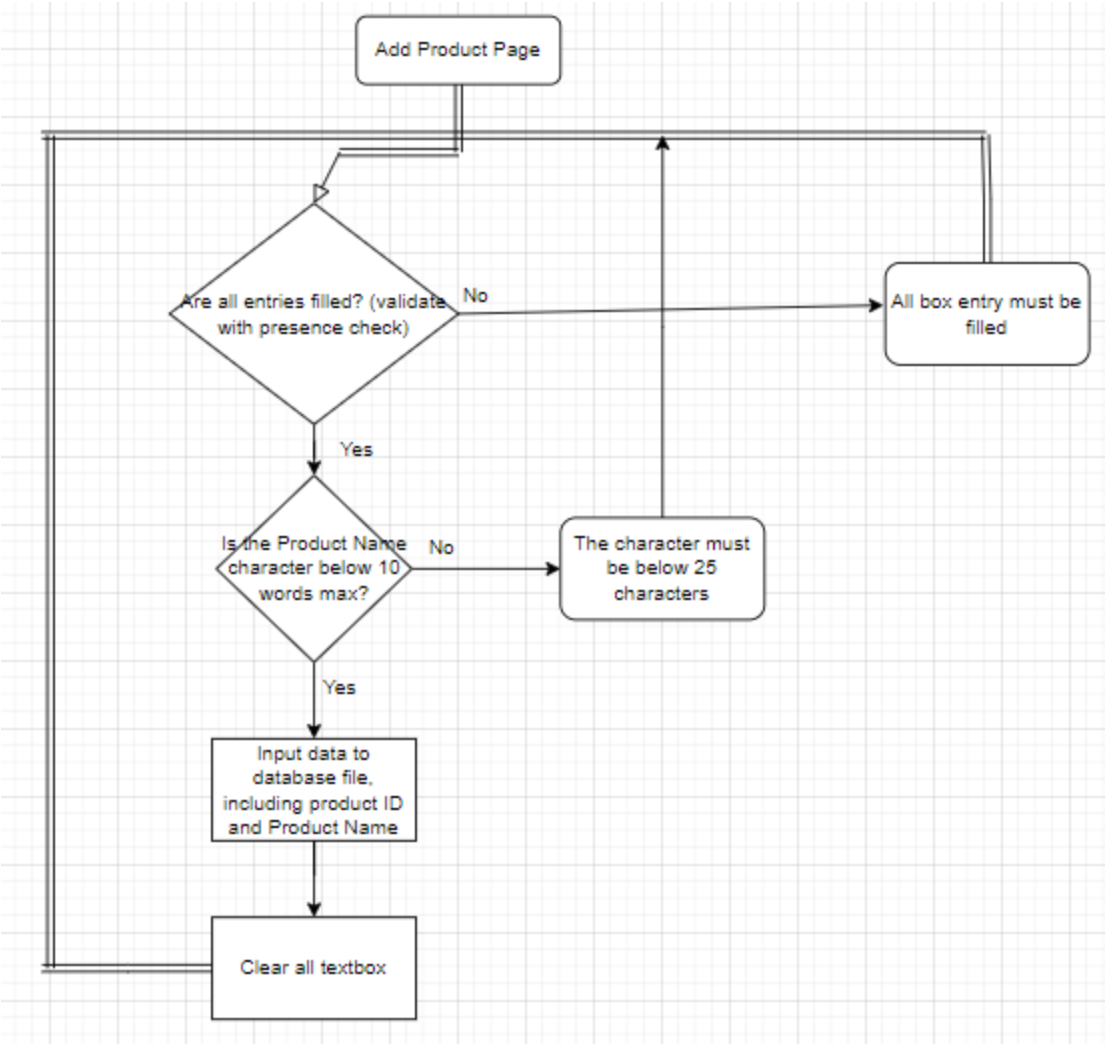

Enter Login Page
User wants to go back to registration page?
user will re-enter registration screen
All box entry must be filled
Fill all boxes entries
Are all entries filled?
Is the username and password listed in the databaase?

Home Page

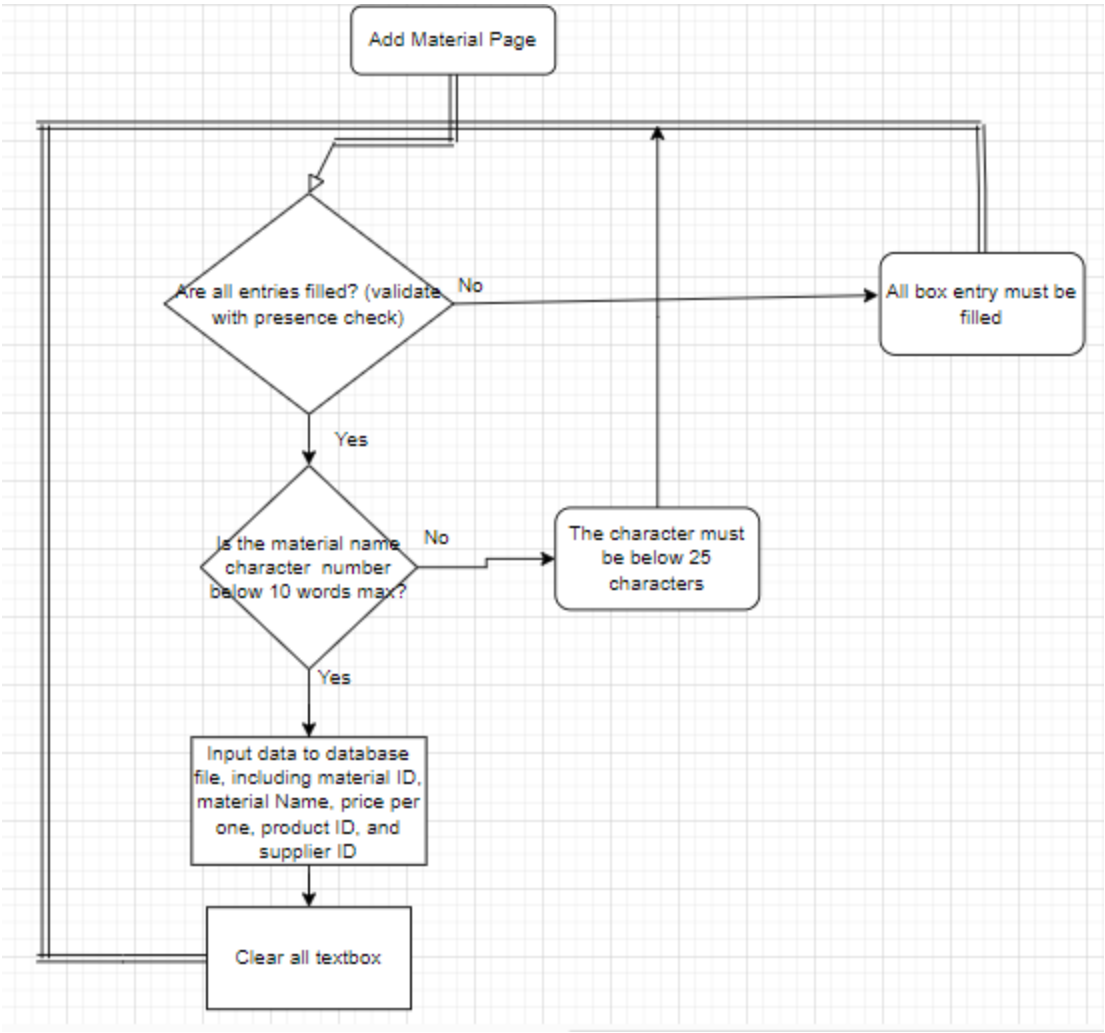

Home Page
Go to "Add Product Page"?
No
User will go to the designated page
Yes
Go to "Add Material Page"?
No
User will go to the designated page
Yes
Go to "Add Supplier Page"?
No
User will go to the designated page
Yes
Go to "Add Customer Page"?
No
User will go to the designated page
Yes
Go to "Add Instalation Page"?
No
User will go to the designated page
Yes



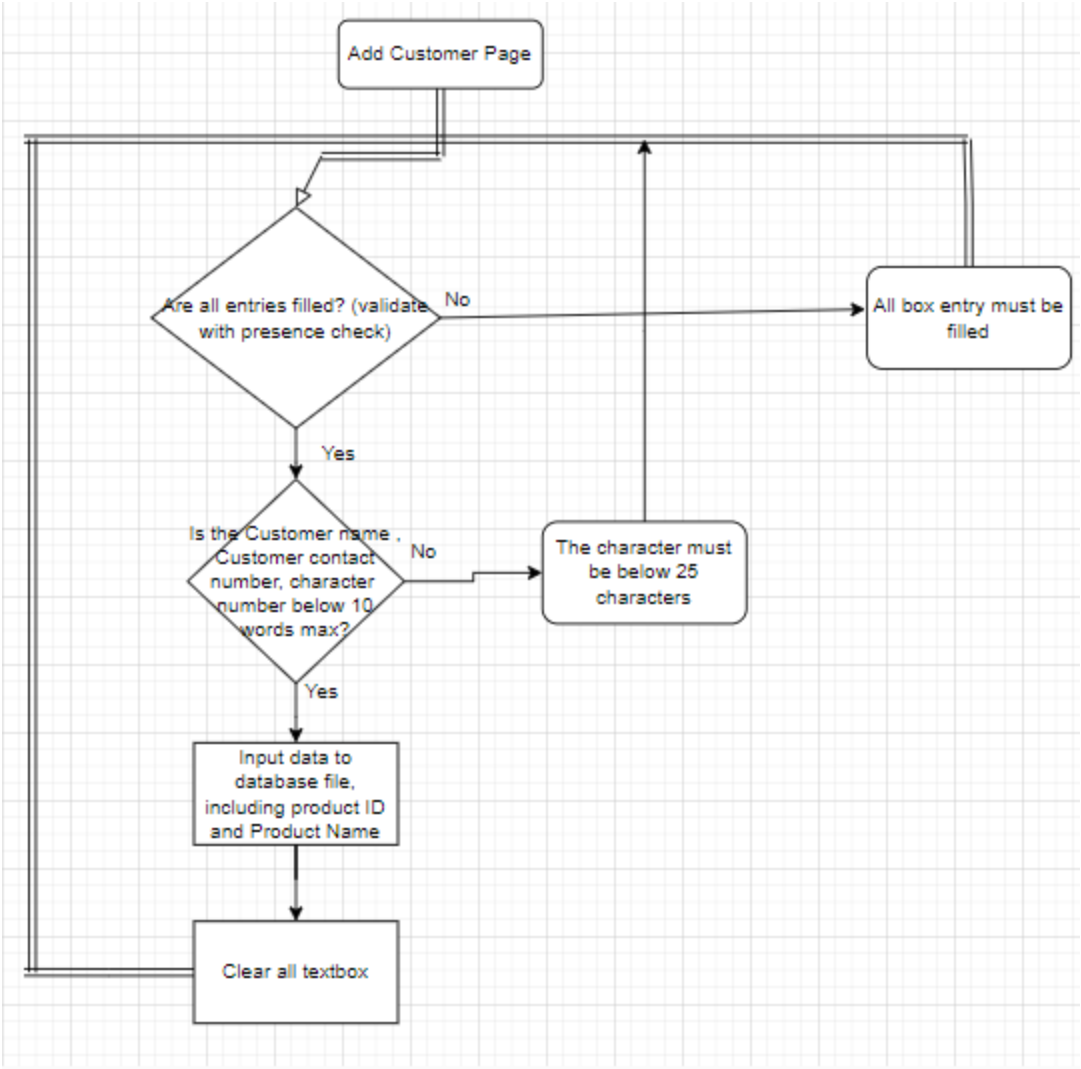
Add Product Page



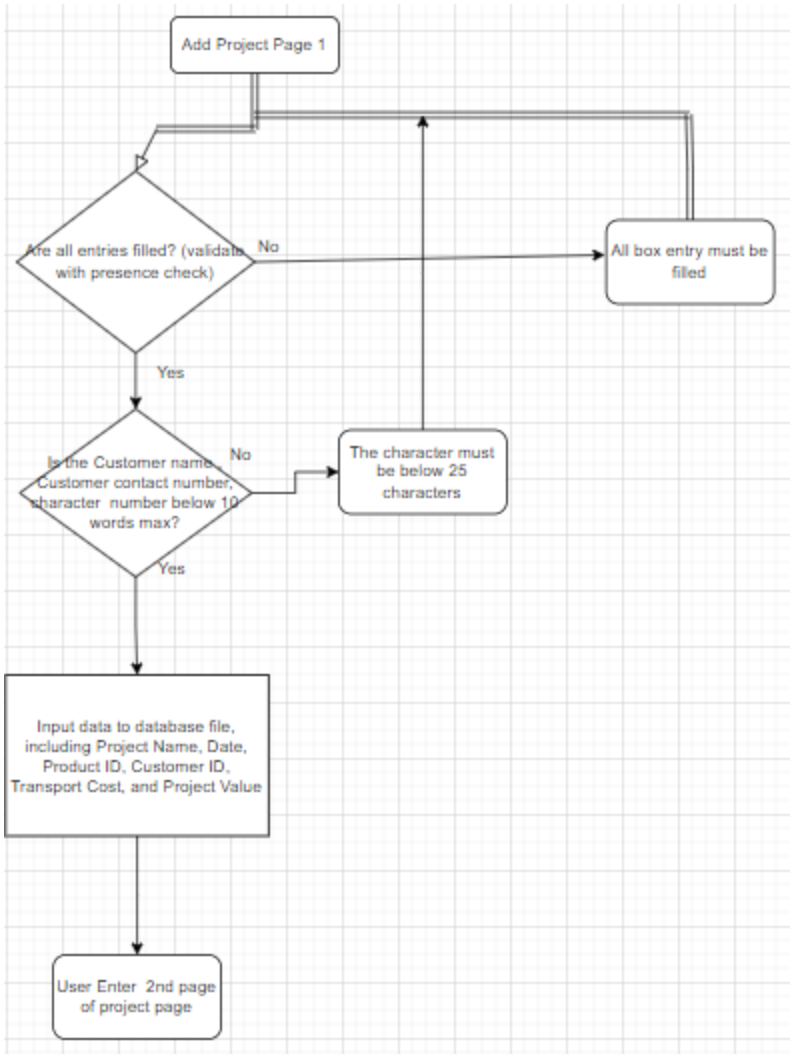
Add material Page



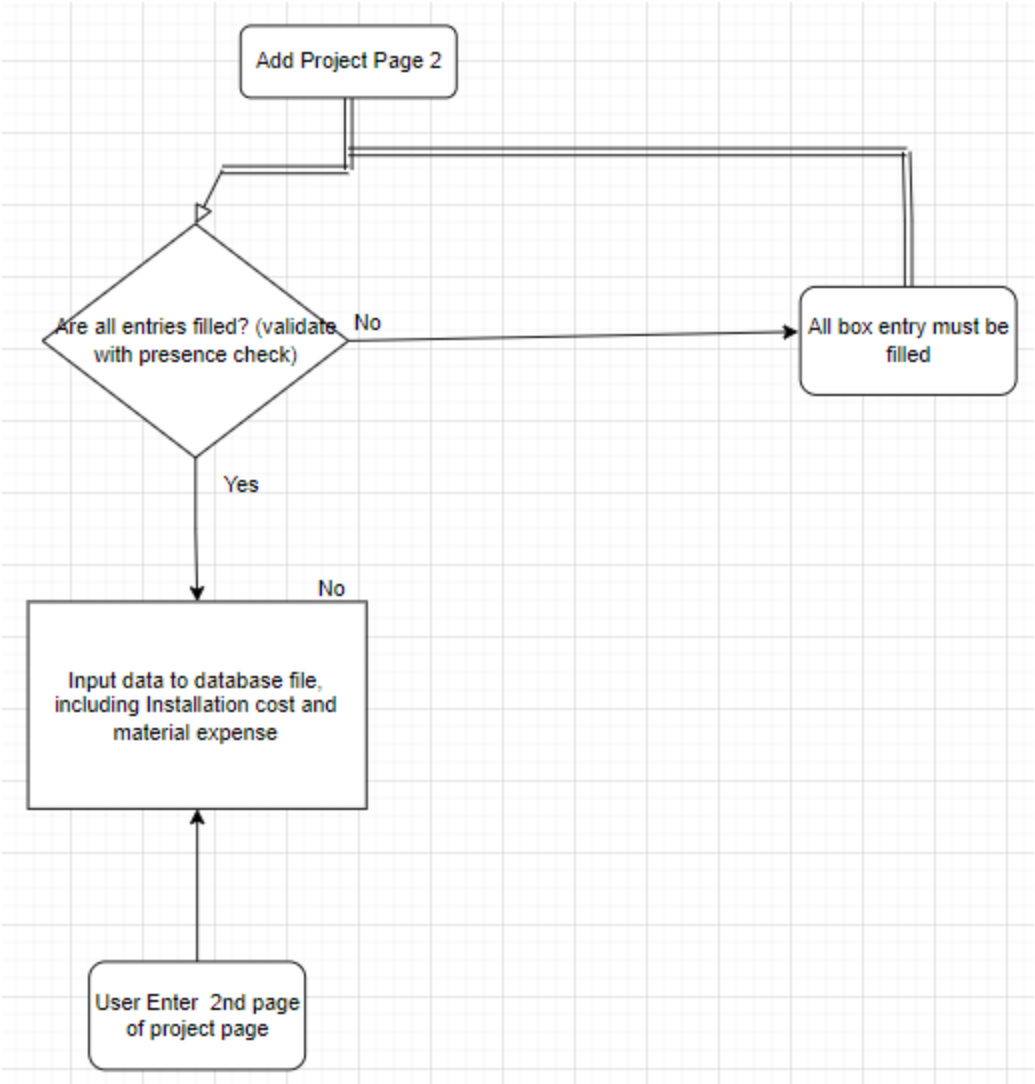
Add customer page



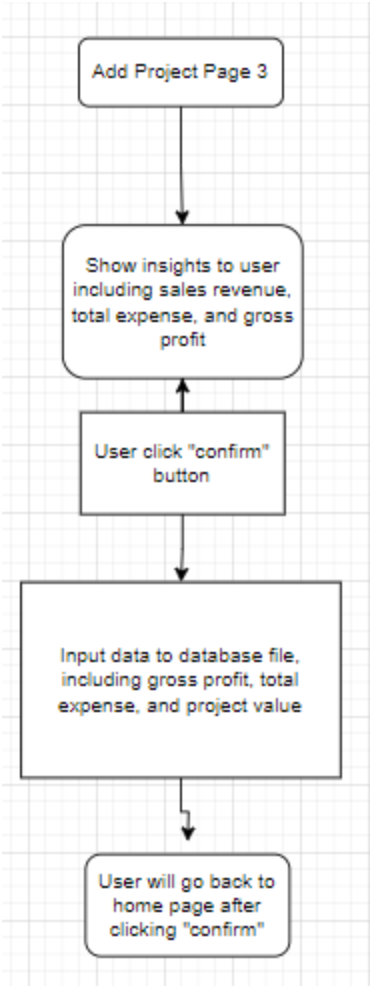
Add Project Page 1



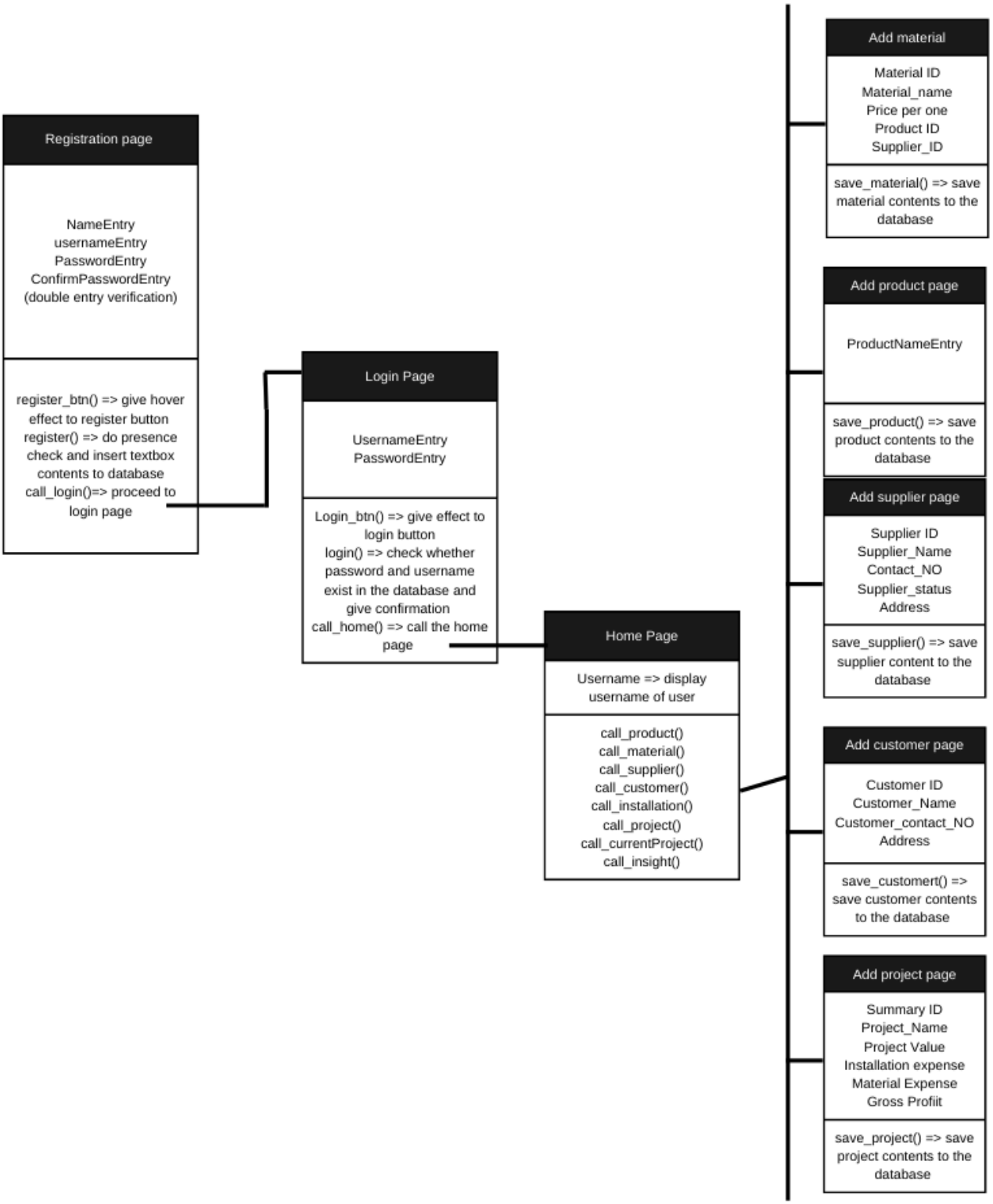
Add Project page 2



Add Project page 3



VII. **Flowchart UML**


Registration page
NameEntry
usernameEntry
PasswordEntry
ConfirmPasswordEntry
(double entry verification)
register_btn() => give hover effect to register button
register() => do presence check and insert textbox contents to database
call_login()=> proceed to login page
Login Page
UsernameEntry
PasswordEntry
Login_btn() => give effect to login button
login() => check whether password and username exist in the database and give confirmation
call_home() => call the home page
Home Page
Username => display username of user
call_product()
call_material()
call_supplier()
call_customer()
call_installation()
call_project()
call_currentProject()
call_insight()
Add material
Material ID
Material_name
Price per one
Product ID
Supplier_ID
save_material() => save material contents to the database
Add product page
ProductNameEntry
save_product() => save product contents to the database
Add supplier page
Supplier ID
Supplier_Name
Contact_NO
Supplier_status
Address
save_supplier() => save supplier content to the database
Add customer page
Customer ID
Customer_Name
Customer_contact_NO
Address
save_customert() => save customer contents to the database
Add project page
Summary ID
Project_Name
Project Value
Installation expense
Material Expense
Gross Profiit
save_project() => save project contents to the database

VIII. Test Plan

Success criteria	Testing method
The solution application must have a register and login system to secure the user's account and avoid unauthorized access to the business data	
The solution application must have page to add product record	Open the product page
The solution application must have page to add material record	Open the material page
The solution application must have page to add suppliers record	Open the suppliers page
The solution application must have page to add installation cost	Open the installation page
The solution application must have page to add customer record	Open the customer page
The solution application must have page to add project record	Open the add project page
The record page should have a screen to manage (add, edit, delete) users of my program	Open the pages and test the manipulation data feature
The solution application should have an integrated database using SQLite that will store all business data from the user and be connected to a cloud database using MYSQL to be able to provide a secure cloud backup as the client requested.	Check and open the databases
The solution application should be able to conduct calculations to gain financial informations regarding the inputted project data such as revenue, loss, and profit of the client company's project	Test the add project page
The solution application should have the feature of showing the client's current ongoing projects	Open the current project page